
PS Analysis in Data Science: Problem Set 8

Problem 1. Consider $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = \sqrt{1 + x^2 + y^2}.$$

Determine all critical points and classify them into local minima, maxima, and saddle points.

Problem 2. Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by

$$f(x, y) = \sin x + \cos y.$$

- Determine all critical points and classify them as local minima, maxima, and saddle points.
- Show that ∇f is L -Lipschitz continuous.
- Compute all critical points within $[0, 2\pi] \times [-\frac{\pi}{2}, \frac{3}{2}\pi]$ in Julia via gradient descent by selecting suitable initial vectors (x_0, y_0) .

Problem 3. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. The Newton iteration is defined by

$$x_{k+1} := x_k - \frac{f(x_k)}{f'(x_k)}, \quad k \in \mathbb{N},$$

whenever $f'(x_k) \neq 0$.

Justify the following interpretation of Newton's method:

In the k -th Newton iteration, the function f is replaced by its affine linear approximation F_k at x_k . The next iterate x_{k+1} is then defined as the solution of the linearized equation $F_k(x) = 0$.

More precisely:

- Determine the affine linear approximation F_k of f at x_k .
- Show that solving $F_k(x) = 0$ yields exactly the Newton iteration formula.

Problem 4. Write a Julia function

`myNewton(f, x0, tol, maxiter)`

that performs Newton's method for a function f , starting from the initial value x_0 . The iteration should terminate when either

$$|x_{k+1} - x_k| < \text{tol},$$

or when the maximal number of iterations `maxiter` is reached. Your function should return:

- the final approximation,
- the number of iterations used,
- and a vector containing all iterates.

Test your implementation on the equations $x^2 - 2 = 0$ and $\cos(x) - x = 0$. Experiment with different initial values x_0 . What do you observe? Plot the convergence history $k \mapsto |x_k - x^*|$ for one suitable example, where x^* denotes the computed root.